E&ICT, IIT Roorkee



Advanced Program in

Applied Data Analytics & Generative AI

By E&ICT Academy, IIT Roorkee

6-Months 140 Hours (Brochure)

About The IIT Roorkee

Indian Institute of Technology Roorkee (IIT Roorkee), one of the oldest technical institutions in Asia, was established in 1847 as the Thomason College of Civil Engineering and was later converted to an IIT in 2001.

The institute offers a wide range of academic programmes including BTech, BArch, BS, MTech, MSc, MBA, MDes, and PhD across its 23 academic departments, centres, and schools covering engineering, sciences, architecture, management, humanities, and social sciences.

Over the years, IIT Roorkee has developed state-of-the-art infrastructure, a strong research ecosystem, and a rich legacy of excellence in technical education and innovation. The institute continues to be recognized nationally and internationally for its academic rigour, research output, and contributions to industry and Society.



About The E&ICT Academy IIT Roorkee

Electronics & ICT Academy, IIT Roorkee, aims to strengthen the skills of faculty members, working professionals, and students from Engineering, Science, Commerce, and Polytechnic institutions through specialized short-term training programs.

The Academy focuses on both fundamental and advanced areas of Information Technology, Electronics & Communication, Product Design, and Manufacturing.

In addition, E&ICT Academy, IIT Roorkee, organizes customized industry-oriented training programs, faculty development programs (FDPs), and research promotion workshops to bridge the gap between academia and industry. These initiatives are designed to enhance technical competency, encourage innovation, and promote collaborative research in emerging technologies.



About The IoT Academy

The IoT Academy, established in 2017 is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. E&ICT Academy, IIT-Roorkee, IIT-Guwahati and IIT-Kanpur for Advance Certification courses to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

30,000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes



About The Program

Dive into the exciting world of Applied Data Analytics and Generative AI. This program helps you learn how to study data, understand it, and use it to make better decisions. You will also learn how generative AI works and how it can create new ideas and solutions. These skills can be used in many fields like healthcare, finance, retail, and technology. This course gives you the knowledge to use data and AI to solve real problems in the real world.



Mastering Data Analytics & Generative AI

- Learn the basics of data analytics. You will understand how to collect data, clean it, and show it in charts. You will also learn simple statistics.
- Generative AI helps by making new data, giving better predictions, and doing some tasks automatically.

Practical Applications

- You will work on real projects. You will explore data, build prediction models, check risks, and test different situations.
- Generative AI will help you find useful insights, work faster, and solve business problems.

Industry Trends & Real-World Impact

- You will study real examples from companies like Google, Microsoft, and IBM.
- You will learn how data analytics and generative AI are changing industries. They help companies make better decisions, give personalized services, and create new ideas.

Ethics, Governance, and Responsible AI

- You will learn about common issues like bias, data privacy, and copyright in AI.
- You will understand how to use AI in a safe and fair way by following rules and protecting data.

Why Learners Choose Us Over the Rest?

Program Perks	Live Learning For Real-Time Experience and Classroom Feel	Other Institution
Online Flexible Learning	✓	✗
1:1 Live Doubt Session	✓	✗
Weekend Classes	✓	✗
Taught by IIT Faculty	✓	✗
Recorded Lectures Lifetime Access	✓	✗
Applied Industry Projects	✓	✓
Live Capstone Projects	✓	✓
Resume & Interview Preparation	✓	✓
2 Month Industrial-Internship	✓	✗
Industry Partner's Alumni Network Access	✓	✗

Why Choose Data Analytics & Generative AI Now?

Market Growth:

The global market for generative AI, a type of technology that can create new content and ideas, is expected to grow rapidly. It's predicted to increase by more than 30% each year from 2024 to 2030, potentially reaching over $100 billion by the end of the decade.

Job Demand:

There is an incredible demand for professionals skilled in artificial intelligence (AI) and machine learning (ML). Job postings for roles related to generative AI have jumped by more than 80% in the past year, indicating that many companies are looking for talent in this area.

Future Readiness:

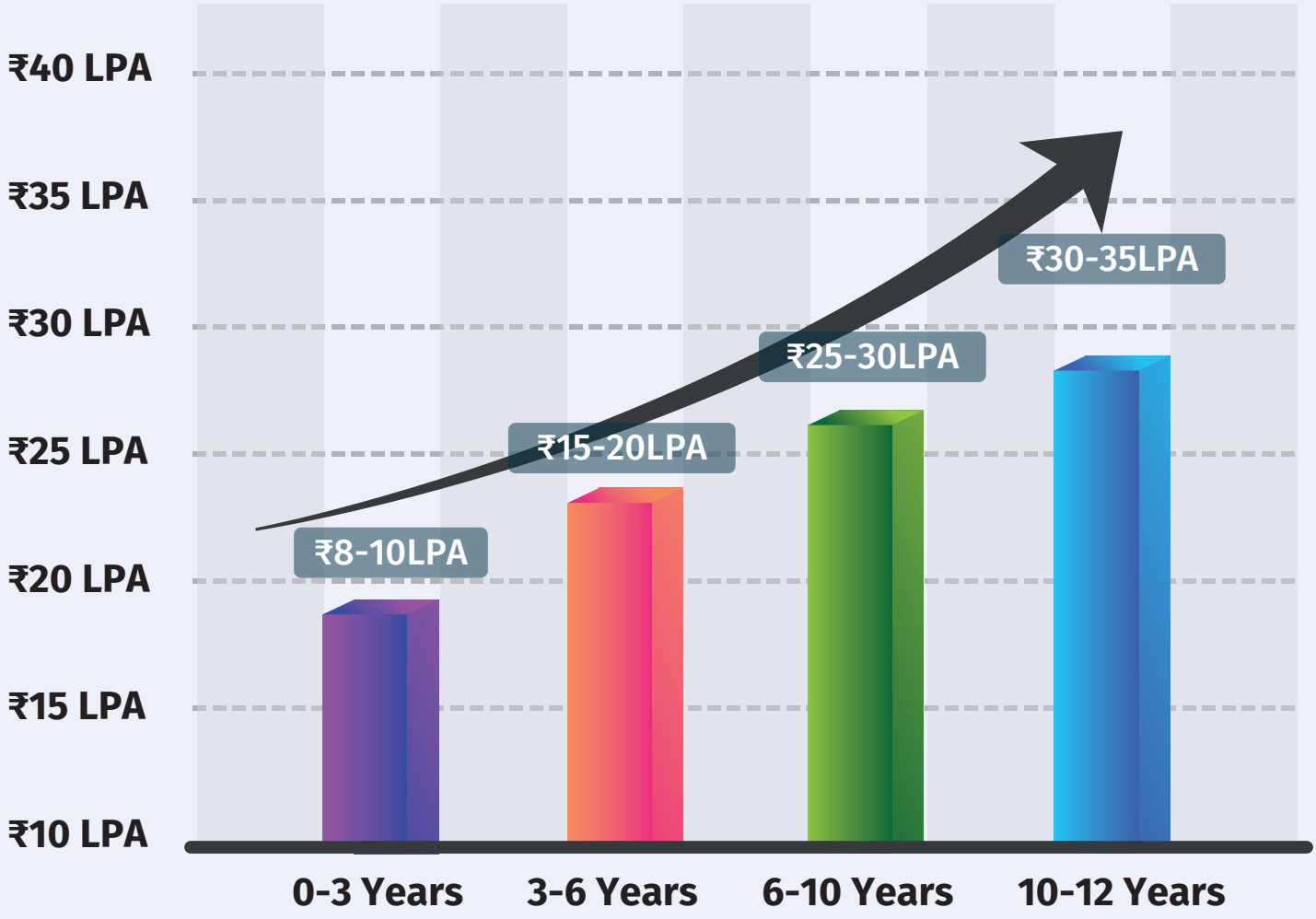
Having skills in generative AI is becoming crucial for various careers, including data science, AI research, software development, and creative fields like design and content creation.

If you're interested in becoming a data scientist, exploring creative professions, or starting your own business, this course will help you utilize the power of generative AI and keep you ahead in a rapidly changing technological world.



Salary & Scope

Median salary of Data Analytics with GenAI Professionals across years of Experience in India



Role/Category	Salary Range (INR	Additional Notes
Large Companies (Senior Roles)	1 crore+ annually	High-end salaries for very experienced professionals.
Startups & Mid-size Firms	30-40 lakh per annum	Varies with experience and domain expertise.
Indian IT Companies (Entry Level)	5-8 lakh per annum	Freshers, typically lower starting salaries.

Role/Category	Salary Range (INR)	Additional Notes
Generative AI Developers	8.5-18 lakh per annum	Mid-level roles with solid skills can earn more.
Non-Tech Generative AI Consultants	21-35 lakh per annum	High starting base salaries with experience.
AI Engineers at Top Tech Firms	10-25 crore range (senior)	Exceptional senior roles at Google, Microsoft, etc.
GenAI Startups	8-24 lakh per annum	Based on skill set and startup growth stage.
ML Engineers at Startups	30-40 lakh per annum	Often higher than average SWE salaries.
Mid-Career Professionals (3-8 yrs)	15-35 lakh per annum	Experienced roles with leadership responsibilities.
Median Salary (Gen AI Professionals)	12-30 lakh per annum	Reflects varying skill levels and companies.
Median Salary (Data Analytics Roles)	6-12 lakh per annum	Core analytics roles without generative AI focus.

Course Fee



eict IIT Roorkee

Advanced Program in Applied Data Analytics & Generative AI

₹ 70,000 + 18% GST

Avail Scholarship Up To 30%

Financial Partner

Candidates can pay the course fee through Netbanking and Credit/Debit cards. Also, with our corporate financial partnerships avail education loans at 0% interest rate

Razorpay PropelId

The IoT Academy
Connecting The Unconnected

Placement Partner

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Payment Partner

What's Included in the Fee

- 140+ Hours of Instructor Led Live Training
- Certification from E&ICT Academy, IIT Roorkee
- Learn From IIT Professors and the Industry Experts
- Access to Course Materials and Resources
- 20+ Hands-on Projects and Case Studies
- 100% Support by the Dedicated Industry Partner Team

Additional Benefits

- Lifetime Access to Live Recorded Sessions
- Complimentary Career Guidance or Resume-building
- 2 Months Internship Experience By Industry Partners
- 1:1 Doubt Resolution In Live Sessions

Tools & Technologies



Admission Process

The application process consists of four simple steps. An offer of admission will be made to the selected candidates and accepted by the candidates upon payment of the admission fee.

1. **Online Application Form**
   Apply by filling a simple online application form You will be required to provide personal, educational, and professional details. Once we receive your details, our Admission Head will reach out to take your candidature further.

2. **Interview Process**
   Go through a screening call with the Admission Director's office who will gauge your passion and eligibility for the program.

3. **Scholarship & Offer Letter (Provided By Industry Partner)**
   Apply for the scholarship (not mandatory) and take the test. An offer letter will be rolled out to the selected candidates.

4. **Admission & Batch Allotment**
   Complete the admission and make a quick block payment formality with assistance from our loan partners, you will be given course credentials and your learning journey will begin!

Introduction to Data Analytics & Generative AI - 3 Hours (1 Session)

- Overview of Data Analytics
- Overview of Artificial Intelligence
- Traditional AI vs. Generative AI
- Evolution of AI models (Rule-Based → ML → DL → Generative AI)
- Industry Use Cases

Data Analytics & Data Visualization with Python and Business Analytics - 45 hours (15 Sessions)

1. **Python Basics**
   - What is Python
   - Application of Python
   - Why use Python for AI-ML Installation
   - Anaconda/Other Idle
   - Python Tokens
   - Data Types in Python
   - Conditional Statement
   - Loops in Python
   - Functions in Python
   - Advance Functions
   - File Handling
   - Mini Project

2. **Intro to Git & Github and Kaggle**
    - Git and GitHub
    - Kaggle
3. **Python NumPy & Pandas**
    - Introduction to NumPy
    - Exploring a NumPy Array
    - Indexing & Slicing a NumPy Array
    - Manipulating a NumPy Array
    - Performing Mathematical & Statistical Functions using NumPy
    - Performing Linear Algebra Operations using NumPy
    - Introduction to Pandas
    - Exploring Pandas Series
    - Introduction to Pandas DataFrame
    - Importing & Exporting Data
    - Implementing basic DataFrame functionalities
    - Exploring Descriptive Statistics with Pandas
4. **Exploratory Data Analysis (EDA)**
    - Data Types
    - Dispersion & Skewness
    - Data imputation
    - Data Pre-processing
    - Data Cleaning
    - Data Manipulation
    - Advanced Manipulation
5. **SQL For Data Analytics**
    - SQL Basic
    - SQL Joins
    - SQL Aggregations
    - Subqueries and Temp Tables
    - SQL Data Cleaning
    - Window Functions

6. **Data Analysis with Excel**
   - Key Formulas and Functions, Ranges and Tables
   - Data Cleaning – Text Functions, Dates, and Times
   - Conditional Formatting, Sorting, and Filtering
   - Dashboard Creation
   - Analysis with Pivot Tables
   - Data Analysis in Excel – Trends and Patterns
   - Data Visualization in Excel – Charts and Plots
   - Functions in Python
   - Advance Functions
   - File Handling
7. **Business Analytics**
   - Business Analysis in a Data + AI World
   - Requirement Gathering for Analytics & AI
   - Documentation Essentials
   - Tools for Business Analytics
   - Process Analysis & KPI Design
   - Data Interpretation & Insight Storytelling
   - Designing GenAI Solutions
   - Agile Business Analysis Essentials
   - Testing & Validation (UAT)
   - Real-World Case Studies Domain Wise
8. **Data Visualization with Python**
   - Why Data Visualization?
   - Introduction to Data Visualization
   - Libraries & Tools for Data Visualization in Python
   - Static Data Visualization Using Seaborn
   - Interactive Data Visualization Using Plotly Express
   - Interactive Animations & Facet Plots

9. **Tableau for Business Intelligence**
    - Understanding Data
    - Creating Your First Visualization
    - Tableau Calculations
    - Formatting Visualizations
    - Manipulating Data in Tableau
    - Creating Dashboards & Stories
    - Distributing & Publishing Your Visualization
10. **Visual Storytelling using Power BI**
    - Understanding Data
    - Creating Your First Visualization
    - Tableau Calculations
    - Formatting Visualizations
    - Manipulating Data in Tableau
    - Creating Dashboards & Stories
    - Distributing & Publishing Your Visualization

Mathematics Fundamentals - 15 hours (4 Sessions)

- Fundamentals of Algebra
    - Vectors, Matrices, Eigenvalues & Eigenvectors
    - Dot Product, Matrix Operations, Applications in AI
- Probability & Statistics
    - Descriptive Statistics (Mean, Median, Mode, Variance, Standard Deviation)
    - Probability Distributions: Normal, Bernoulli, Binomial
    - Central Limit Theorem
- Fundamentals of Calculus

Module 1 Module 2 Module 3 **Module 4** Module 5 Module 6 Module 7
Module 8 Module 9 Module 10

NLP Foundation - 15 hours (5 Sessions)

- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)
- Recurrent Neural Network (RNN)

Module 1 Module 2 Module 3 Module 4 **Module 5** Module 6 Module 7
Module 8 Module 9 Module 10

Foundations of Generative AI & Prompt Engineering - 12 Hours (4 Sessions)

Introduction to Generative AI

- What is Generative AI? Evolution from Rule-Based AI to GenAI
- Differences: Generative AI vs. Traditional AI
- Industry Applications:
  - Content Creation (Chatbots, Story Writing, Blogging)
  - Code Generation (AI-Assisted Programming)
  - Image & Video Synthesis (DALL·E, MidJourney, DeepSeek Vision)

Prompt Engineering & Optimization

- Fundamentals of Prompt Engineering
  - Zero-shot, Few-shot, Chain-of-Thought (CoT) Prompting
  - Role-based Prompts, Multi-turn Conversations
- Advanced Prompting Techniques
  - System vs. User Prompts, Style Transfer, and Iterative Prompting

Generative AI Architectures & Large Language Models (LLMs) - 24 Hours (8 Sessions)

Encoder-Decoder Architectures & Attention Mechanism

- Understanding Sequence-to-Sequence (Seq2Seq) Models
- Introduction to Attention Mechanisms
  - Self-Attention & Cross-Attention
  - Scaled Dot-Product Attention & Multi-Head Attention
- Transformers & Evolution
  - Introduction to Transformers
  - Positional Encoding & Feed-Forward Networks

Foundation of Generative Models

- Types of Generative Models
  - Variational Autoencoders (VAEs)
  - Generative Adversarial Networks (GANs) – StyleGAN, CycleGAN
  - Transformer-Based Models (GPT, BERT, T5)
- Understanding AI Hallucinations, Bias, and Model Safety
  - Variational Autoencoders (VAEs)
  - Generative Adversarial Networks (GANs) – StyleGAN, CycleGAN
  - Transformer-Based Models (GPT, BERT, T5)

Large Language Models (LLMs) & Fine-Tuning

- Popular LLMs: GPT-4, DeepSeek, LLaMA, Mistral, Claude, Falcon
- Fine-Tuning vs. Pre-training of LLMs
  - Transfer Learning for LLMs
  - Customizing LLMs for Domain-Specific Use Cases

Retrieval-Augmented Generation (RAG) & Multimodal AI - 18 hours (6 Sessions)

Retrieval-Augmented Generation (RAG)

- Why RAG? Understanding LLM Limitations
- RAG Pipeline Components:
  - Document Embeddings & Vector Databases (Pinecone, ChromaDB)
  - Query Processing & Context Retrieval

Foundation of Generative Models

- Implementing RAG with Hugging Face & LangChain
- What is LlamaIndex?
- Using LlamaIndex for Efficient Document Retrieval

Multimodal AI – Text, Image, Video & Speech Processing

- Text-to-Image & Image-to-Image Generation
  - How DALL·E, MidJourney, and Stable Diffusion Work
  - Latent Diffusion Models (LDMs) & Hugging Face Implementations
- Speech-to-Text & AI Voice Processing
  - Whisper, Wav2Vec, DeepSeek-STT, Real-Time Speech Transcription
- Advanced Multimodal Models
  - OpenAI GPT-4V, Gemini
  - Text-to-Image, Image Captioning

Agentic AI & Hugging Face Ecosystem - 15 Hours (5 Sessions)

Agentic AI & Autonomous AI Agents

- What is Agentic AI?
  - Understanding Autonomous AI Systems
- AI Agent Frameworks:
  - LangChain, AutoGPT, Hugging Face Agents
  - Building AI Agents for Research, Trading, and Business Automation

Hugging Face Ecosystem

- Hugging Face Models, Spaces, and Transformers
- Fine-Tuning LLMs & Custom AI Deployments
- Enterprise Use Cases
  - AI Chatbots, Code Generation, AI for Research

Real-Time Generative AI in Business Applications

- Generative AI in Healthcare
  - AI-Powered Drug Discovery, Medical Imaging
- AI in Finance
  - Synthetic Data for Risk Modeling & Fraud Detection
- Automating Business Processes with AI

Module 1 Module 2 Module 3 Module 4 Module 5 Module 6 Module 7
Module 8 **Module 9** Module 10

Future of AI & Capstone Project - 5 Hours (2 Sessions)

Ethics, AI Governance & Future Trends

- Responsible AI & AI Governance
  - EU AI Act, OpenAI's AI Policy
- The Future of Generative AI
  - AI-Powered Autonomous Systems
  - Trends in AGI (Artificial General Intelligence)

Module 1 Module 2 Module 3 Module 4 Module 5 Module 6 Module 7
Module 8 Module 9 **Module 10**

Select A Domain That Aligns With Your Goals

- Generative AI in Marketing
- Generative AI in Healthcare
- Generative AI in Legal Services
- Generative AI in Retail and E-Commerce
- Generative AI in Finance
- Generative AI in Education
- Generative AI in Human Resources (HR)
- Generative AI for Leadership Roles

Certification

Upon successfully completing this program, you'll earn certificate in Advanced Program in Applied Data Analytics & Generative AI that is co-branded by E&ICT Academy, IIT Roorkee.

This certificate will testify to your skills as an expert in Data Analytics & Generative AI

SAMPLE CERTIFICATE





Live Industry-Based Projects

Industry projects are an important part of the Advanced Program in Applied Data Analytics & Generative AI. These projects let you work on real problems from real companies. You get hands-on experience in data analytics and generative AI. This helps you learn practical skills and solve tough problems in different industries.

Text-to-Image Generation

Description: Combine the power of NLP and CV to generate realistic images from textual descriptions, such as creating visual prototypes for design concepts or generating fictional landscapes.

Tools and Technologies: PyTorch, Tensor-Flow, OpenCV, Diffusion Models, GANs, Transformers.

Outcomes: Learn to preprocess text and image data, design transformer-based models, and implement diffusion models for image synthesis.





Interactive Chatbot for Healthcare

Description: Build a conversational agent using GPT-based LLMs that assists users with healthcare queries, such as symptom checks or medication reminders.

Tools and Technologies: Rasa, PyTorch, Hugging Face Transformers, Streamlit, NLP Libraries (spaCy, NLTK).

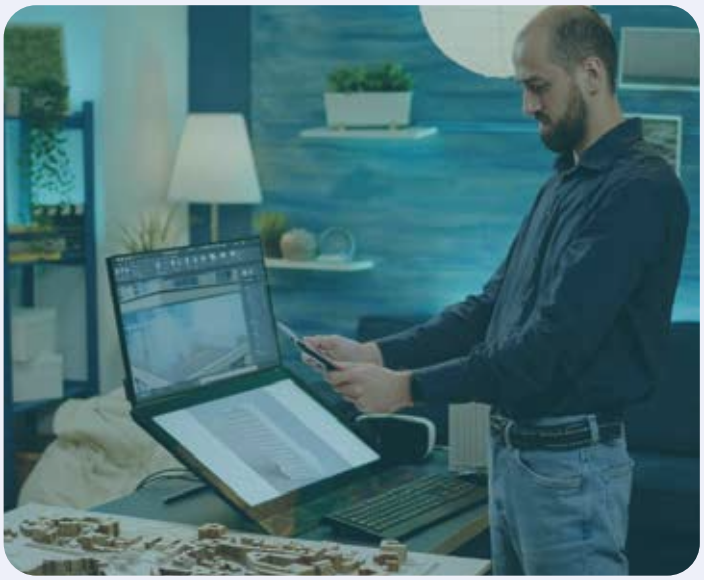
Outcomes: Gain experience fine-tuning LLMs for domain-specific tasks, handling conversational context, and deploying chatbots.

Generative Art Creation

Description: Develop a system to generate artwork using GANs or style transfer techniques. The model will create unique artistic styles or merge different styles into new creations.

Tools and Technologies: TensorFlow, PyTorch, StyleGAN, OpenCV.

Outcomes: Understand GAN architecture, apply advanced loss functions, and explore applications in digital art and design.





Predictive Maintenance with Generative Models

Description: Use Variational Autoencoders (VAEs) to detect anomalies in industrial sensor data and predict maintenance needs.

Tools and Technologies: PyTorch, TensorFlow, VAE, Pandas, NumPy.

Outcomes: Learn latent space representation, anomaly detection, and how generative AI can enhance predictive maintenance.

AI-Powered Content Generation for Marketing

Description: Create a generative AI model to assist marketing teams in generating product descriptions, blog posts, and ad copies.

Tools and Technologies: GPT, Hugging Face, PyTorch, TensorFlow.

Outcomes: Work with fine-tuning strategies for LLMs, data cleaning, and evaluating text quality using BLEU and ROUGE metrics.





Viral Content Intelligence System

Description: Analyze social media content to identify viral patterns and predict high-performing posts using engagement and trend signals.

Tools and Technologies: Python, NLP, Social Media APIs, Machine Learning, Power BI / Tableau.

Outcomes: Understand content virality factors, build trend-detection models, and optimize content strategies.

Voice Cloning and Audio Synthesis

Description: Develop a model to synthesize human-like speech or clone specific voices using generative models like WaveNet or Tacotron.

Tools and Technologies: TensorFlow, PyTorch, Librosa, WaveNet.

Outcomes: Learn to preprocess audio data, generate waveforms, and build real-time voice applications.





AI-Powered Story Writing

Description: Create an LLM-based application for collaborative storytelling, where users provide input, and the model generates the next part of the story.

Tools and Technologies: GPT, Hugging Face, Streamlit, PyTorch.

Outcomes: Implement LLMs for creative writing, build interactive web apps, and handle user-driven input generation.

Sentiment Analysis of Customer Feedback

Description: You will use NLP and machine learning to study customer comments. The system will mark each comment as positive, negative, or neutral. This helps you understand customer feelings and what they like or dislike.

Tools: Python (NLTK, spaCy, scikit-learn, BERT), TensorFlow, and libraries for charts.

Outcomes: You will clean the text, build a sentiment model, and create charts to show trends. You will also learn how to use the results to improve products or services.





Visualization of Life Expectancy and GDP Over Time

Description: You will study how life expectancy and GDP change across countries and years. You will make interactive charts to show patterns and differences.

Tools: Python (pandas, matplotlib, seaborn, Plotly), Jupyter Notebook, Excel, and public datasets.

Outcomes: You will learn how to clean data, combine datasets, and tell stories with charts. You will make useful plots like scatter, line, and animated charts.

Analyze Sales Dataset with Pivot Tables & Charts

Description: Summarize and visualize sales data to discover insights like trends, category-wise performance, and regional sales using Excel's Pivot Tables and charts.

Tools and Technologies: Microsoft Excel, Pivot Tables, Pivot Charts

Outcomes: Learn to structure raw sales data, create summaries, and visualize using pie, bar, column, and line charts for actionable business intelligence.





Customer Dataset Cleaning & Analysis

Description: Cleanse customer records to remove duplicates, fix data format issues, and fill missing values, ensuring robust analysis and reliable business decisions.

Tools and Technologies: Excel formulas, Python (pandas), SQL

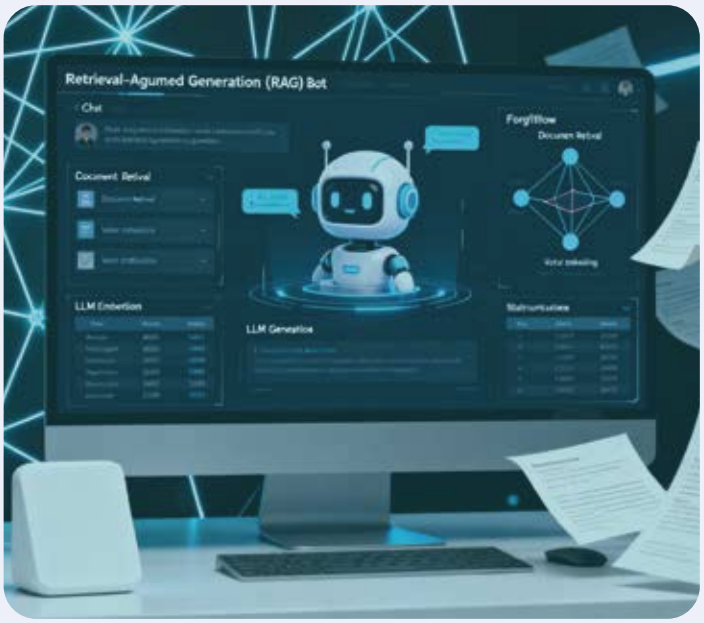
Outcomes: Master systematic approaches for identifying errors, standardizing data fields, and validating customer information for downstream analysis.

Explore Sales Dataset Distribution

Description: Analyze the spread of sales figures using statistical measures and distribution plots to identify outliers and assess sales variability.

Tools/Technologies: Excel, Python (matplotlib, seaborn), Histograms, Box Plots

Outcomes: Develop the ability to compute summary statistics, generate distribution charts, and detect anomalies or patterns within sales data.





RAG-Based Business Q&A Bot

Description: Create a Retrieval-Augmented Generation (RAG) bot that answers business questions using both structured and unstructured company data.

Tools/Technologies: Python, Transformers, OpenAI/LLMs, Retrieval pipeline (Haystack, LangChain)

Outcomes: Understand text preprocessing, transformer-based retrieval, and generation architectures to enable informed business Q&A and analytics.

Autonomous Business Research AI Agent

Description: Create an autonomous AI agent that conducts market and competitor research by gathering, analyzing, and summarizing business data.

Tools and Technologies: LLMs, LangChain, Web Data Extraction, Python, Vector Databases.

Outcomes: Learn to build autonomous agents for market intelligence and strategic research.





AI Trading & Portfolio Monitoring Agent

Description: Design an AI-driven system to analyze market data, track portfolios, and generate trading insights using predictive models.

Tools and Technologies: Time-Series Analysis, Machine Learning, Python, Financial APIs, Data Visualization.

Outcomes: Understand algorithmic trading basics, portfolio analytics, and real-time monitoring systems.

Customer Growth Intelligence Dashboard

Description: Analyze customer acquisition, retention, and churn patterns using data-driven dashboards to identify growth opportunities and optimize business strategies.

Tools and Technologies: Python, Pandas, SQL, Power BI / Tableau, Machine Learning, Scikit-learn.

Outcomes: Learn to build interactive dashboards, perform customer segmentation, and generate actionable growth insights.





Startup Financial Health & Burn-Rate Radar

Description: Monitor startup financial metrics such as burn rate, runway, and profitability to assess financial sustainability and future risk.

Tools and Technologies: Python, Excel, SQL, Financial Modeling, Power BI, Machine Learning.

Outcomes: Gain experience in financial analysis, forecasting burn rate, and building investor-ready financial dashboards.

Business Risk Probability Simulator

Description: Simulate business risk scenarios by analyzing market, operational, and financial data to estimate risk probabilities.

Tools and Technologies: Python, Statistical Modeling, Monte Carlo Simulation, Machine Learning, Data Visualization.

Outcomes: Learn probabilistic modeling, risk assessment techniques, and scenario-based business forecasting.





Prompt Engineering Studio for Business Automation

Description: Design and test optimized prompts to automate business workflows such as reporting, customer support, and content generation using LLMs.

Tools and Technologies: OpenAI API, LangChain, Prompt Engineering Techniques, Python, LLMs.

Outcomes: Master prompt optimization, build reusable prompt templates, and automate real-world business tasks.

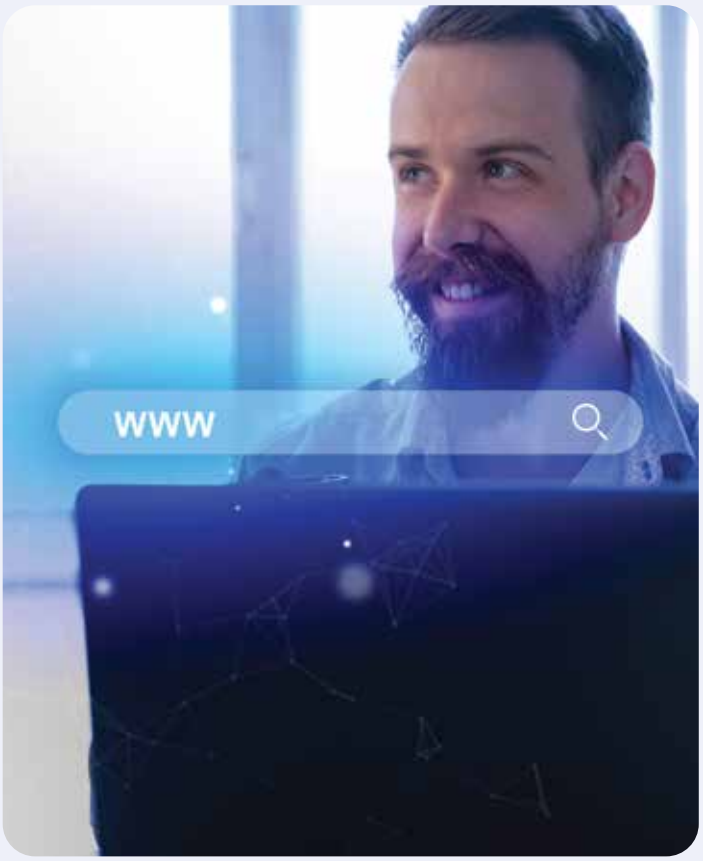
AI Code Assistant for Data Analysts

Description: Build an AI-powered assistant that helps data analysts write, debug, and optimize SQL and Python code.

Tools and Technologies: LLMs, Python, SQL, LangChain, Vector Databases, Prompt Engineering.

Outcomes: Learn to create intelligent coding assistants and improve analytics productivity using AI.





Domain-Specific LLM Fine-Tuning System

Description: Fine-tune large language models on domain-specific datasets to improve accuracy and relevance for specialized use cases.

Tools and Technologies: Hugging Face, PyTorch, Transformers, Fine-Tuning Techniques, GPUs.

Outcomes: Gain hands-on experience in LLM fine-tuning, dataset preparation, and domain adaptation.

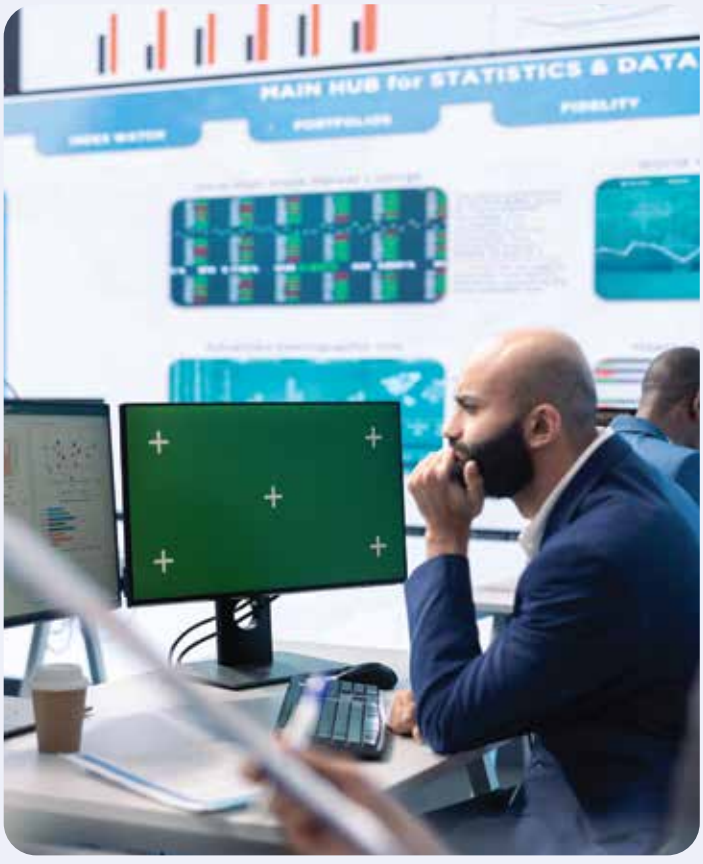
Autonomous Research Paper Summarization LLM

Description: Develop an autonomous system that reads, understands, and summarizes research papers into concise insights.

Tools and Technologies: NLP, Transformers, LLMs, PDF Parsing, Python, LangChain.

Outcomes: Learn document understanding, abstractive summarization, and building research automation tools.





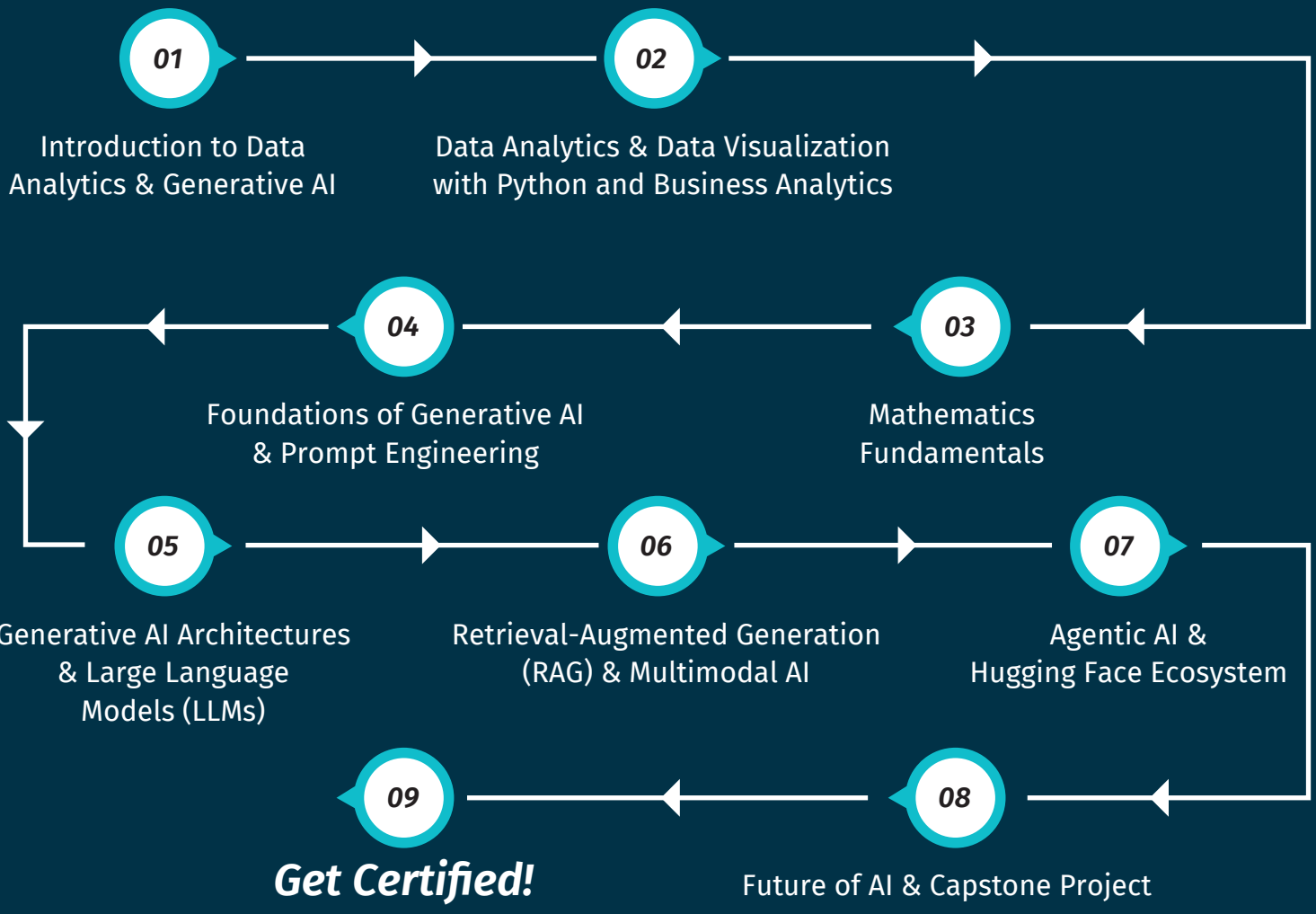
LLM-Powered Contract Risk Analyzer

Description: Analyze legal contracts using LLMs to detect risk clauses, compliance issues, and potential liabilities.

Tools and Technologies: NLP, LLMs, Legal Text Analysis, Python, Vector Databases.

Outcomes: Understand legal document intelligence, risk extraction, and AI-driven contract analysis.

Learning Path



100% Dedicated Job Support By Industry Partner (The IoT Academy)

Live Career-Oriented Webinars

Live webinar sessions that include curriculum and career services walk through to help learners understandt heir learning objective and expectations of hiring managers.

Leadership Skill Development Sessions

Recurring training sessions with experts to help learners develop Interpersonal Leadership Skills.

1-on-1 Career Mentoring Sessions

One-on-one Career Mentoring sessions on how to develop the right skills and attitudeto securea dream job

Exhaustive Interview Preparation

Experttips,sample interview questions, mock interviews with constructive feedback from industry expertsto gain handson experienceof technical rounds, HR round, and more.

Job Search Assistance & Job Feeds

Access to multiple job portals to help learners navigate through thousands of jobs including global remote jobs.

Profile Building Assistance

A dedicated Career Coach will provide experttips on how to create an attractive, relevant resume and LinkedIn profile.

Learning Outcomes

Core Knowledge

Understanding Data Analytics & Generative AI

Build a strong foundation in data analytics principles, Python-based analysis, and generative AI concepts. Distinguish between descriptive, predictive analytics, and generative models.

Applications in Analytics & AI

Explore real-world use cases such as business intelligence, automated reporting, synthetic data generation, text/image creation, and decision support systems.

Types of Analytical & Generative Models

Understand and implement statistical models, regression, classification, clustering, as well as GANs, VAEs, Diffusion Models, and transformer-based architectures.

Technical Skills

Data Analysis & Model Development

Learn to analyze, visualize, and interpret datasets; build, train, and fine-tune machine learning and generative AI models using frameworks like Python, TensorFlow, and PyTorch.

Data Handling & Processing

Master data preprocessing, feature engineering, augmentation, and working with large real-world datasets for analytics and model training.

Model Evaluation & Deployment

Use industry-standard metrics and techniques to evaluate the performance of analytical and generative models (e.g., accuracy, F1 score, FID) and deploy them in enterprise environments.

Problem-Solving and Creativity



Innovative Solutions

- Leverage generative AI to solve creative and complex problems in fields like gaming, marketing, and product design.

Ethical and Responsible AI

- Address ethical challenges such as bias, misuse, and implications of AI-generated content.

Practical Implementation



Deployment of Generative Models

- Learn techniques to deploy and scale generative AI solutions in real-world scenarios.

Hands-On Projects

- Gain experience through projects like creating chatbots, generating realistic images, or synthesizing audio.

Future Readiness



Stay Updated

- Be equipped to understand advancements in generative AI, including new model architectures and techniques.

Career Opportunities

- Open pathways to careers in AI research, machine learning engineering, creative technology, and innovation design.

Who can do this course?

Here's a list of who might benefit from or be eligible to take a Generative AI course:



Students and Academics

- Computer science or engineering students.
- Researchers in AI, machine learning, or related fields.
- Those pursuing postgraduate studies in AI or data science.

Tech Professionals

- Software developers looking to integrate AI tools.
- Data scientists and analysts.
- AI/ML engineers seeking to deepen their expertise.
- IT professionals exploring AI applications.

Business Professionals

- Entrepreneurs leveraging AI for startups.
- Product managers in tech-driven companies.
- Business analysts interested in AI-driven insights.

Creative Professionals

- Digital artists exploring AI-generated art or design.
- Writers and content creators interested in AI-driven tools.
- Game developers using AI for design and storytelling.

Industry-Specific Professionals

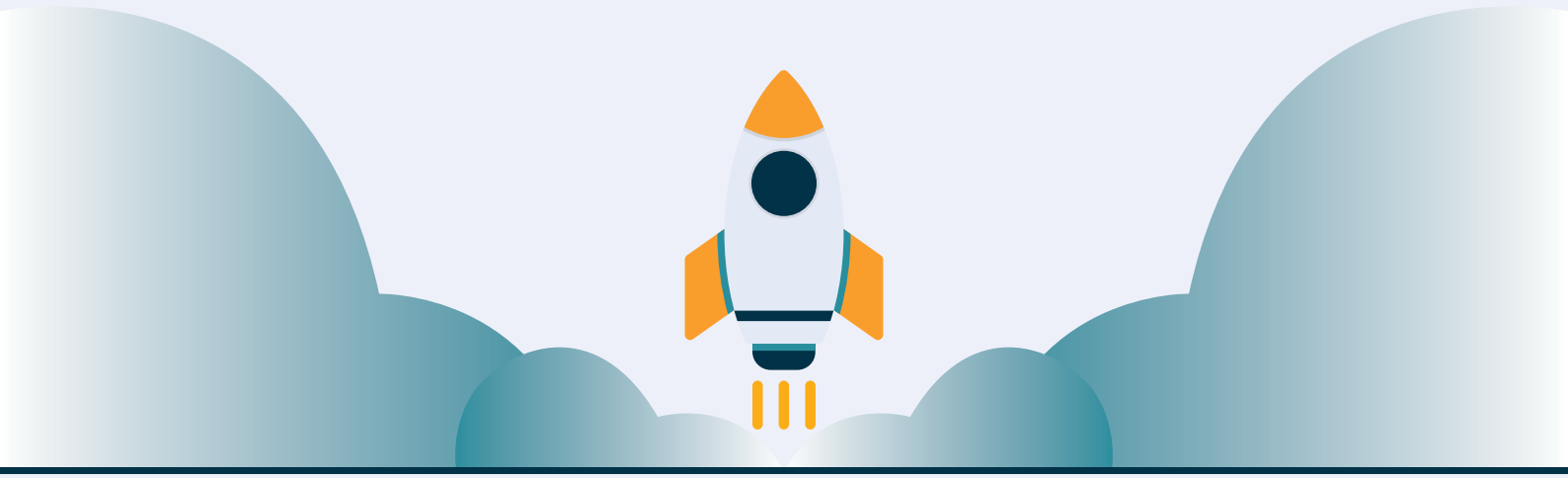
- Software developers looking to integrate AI tools.
- Data scientists and analysts.
- AI/ML engineers seeking to deepen their expertise.
- IT professionals exploring AI applications.

General Enthusiasts

- Anyone passionate about AI and its creative potential.
- Hobbyists who enjoy exploring cutting-edge technology.

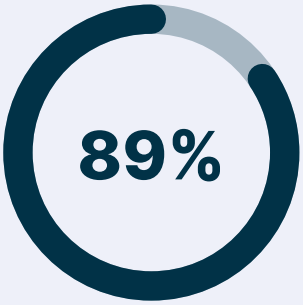
Prerequisite Knowledge

- Familiarity with programming (Python is common).
- Basic understanding of machine learning concepts is often recommended.

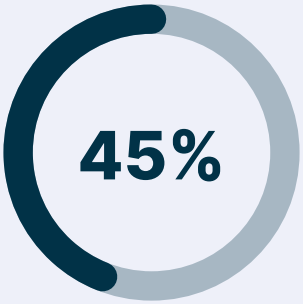


Placement Statistics

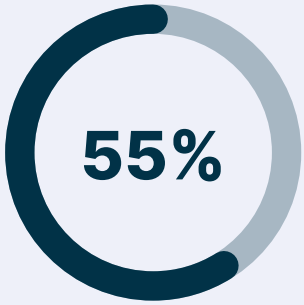
Average Salary	Average Hike
9LPA	60%



Placement Ratio



Career Transition



Fresher Students

Corporates Who Hired form Us

HSBC, KPMG, IBM, wipro, ORACLE
genpact, Moody's Analytics, accenture, kotak Kotak Mahindra Bank, Honeywell
firstsource, HDFC BANK We understand your world, ICICI Bank, WNS, Infosys
Deloitte., BOSCH, EY Building a better working world, tcs, IKEA
cult.fit, Directi Intelligent People. Uncommon Ideas., RIPPLING, MindTickle, amazon

400+ Partner Companies

What The IoT Academy's Learners Say

Abhinav

Google

An exceptional hands-on experience! The Applied Data Science course by The IoT Academy in collaboration with IITG provided me with good project exposure that truly prepared me for the industry. With expert guidance and a collaborative community.

Raghvendra Tiwari

Google

The digital marketing course really impressed me! It taught me about search engine optimization (SEO), social media strategies, and how to analyze data giving me the tools I need to help any brand succeed online. This course is perfect for marketers at any level!

Deep Acharjee

Google

I completed a 45-day industrial internship with IoT Academy in an online format. The entire team at the academy was exceptional. Additionally, the sales team provided all the necessary information and support to ensure the smooth operation of my internship throughout its duration. Sir Devesh, an outstanding faculty member, offered in-depth technical insights into embedded systems and IoT, addressing all of my queries. I am deeply grateful to the entire IoT Academy team for their assistance and guidance.

Ajitesh Rana

Google

I recently had the pleasure of enrolling in IoT and Python courses at The IoT Academy, and I am genuinely impressed with the quality and depth of the content offered. From start to finish, the experience was nothing short of exceptional. The IoT Academy's IoT and Python courses exceeded my expectations on all fronts. If you're looking to gain a solid understanding of IoT concepts and harness the power of Python in IoT applications, I wholeheartedly recommend enrolling in their courses.

Vitoka H Sema

Google

It's a very informative and practical place to learn. The I highly recommend others looking for a course in IoT and Embedded system and even other courses the faculties are very friendly and fun loving and their way of teaching is very practical. I learned a lot of new things.

Ankit Kumar

Google

A wonderful learning experience with the Iot academy, everything is good from curriculum to trainers so supportive.

Sana Tasneem

Google

The Iot Academy provided a wonderful learning platform and experience for the students, teached by an amazing faculty. Every topic was covered thoroughly and made easy for students to understand.

RAVI KASAUDHAN

Google

It's the best platform of learning The IoT Academy, You can easily learn online Internet of things .... Follow their official site & also you can follow on Instagram and LinkedIn

Mayank Bhandari

Google

I think the experience with IoT Academy was amazing. I was provided with current problem of COVID-19 and was asked to visualize and predict the scenario, which i really loved to work upon, and meantime learned a lot during doing that. I even got certificate after successful completion of work. So at the end, i will just say i enjoyed it.

Anmol Pruthi

Google

I was not sure about doing the Java course but when I joined the IoT Academy, I saw a great change in myself. I was getting more eager to learn everything from the trainer. The best atmosphere I have seen of learning is at The IoT Academy. If you want to get best out of you Join the IoT Academy.

Goldie Behl

Google

I had a great time doing this course and everyone involved in The IoT Academy has made it a great experience. I have been recommending you to everyone I know. Thank you for all the assistance and feedback, it has been delightful and very gratifying.

Harshita Mahanta

Google

The training was really awesome, it helped me to gain new knowledge not only about IoT but also about the importance of programming. The experts, guides,mentors all were very supportive and provided much valuable advices.The contents that were provided included all the necessary things and really helped me to learn something new.

E&ICT, IIT Roorkee

Advanced
Program in Applied Data Analytics & Generative AI

By E&ICT Academy, IIT Roorkee

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